

Program 7:

Project Structure:

Pr7

-> nginx-deployment.yaml

Step 1: Start Minikube

minikube start

Step 2: Create Deployment file

```
! nginx-deployment.yaml X
! nginx-deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: nginx
10   template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16       - name: nginx
17         image: nginx:latest
18         ports:
19         - containerPort: 80
```

Step 3: Apply the nginx-deployment.yaml

```
kunaltiwari@riot pr8 % kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 4: Check Status

```
kunaltiwari@riot pr8 % kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	0/2	2	0	12s

```
kunaltiwari@riot pr8 % kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-hlpn2	0/1	ContainerCreating	0	20s
nginx-deployment-6f9664446b-vdxfw	0/1	ContainerCreating	0	20s

Step 5: Expose Deployment as a Service

```
kunaltiwari@riot pr8 % kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80

service/nginx-deployment exposed
```

Step 6: Check Service

```
kunaltiwari@riot pr8 % kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	3m37s
nginx-deployment	NodePort	10.97.9.209	<none>	80:32767/TCP	8s

Step 7: Access the Web App

```
kunaltiwari@riot pr8 % minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32767

🚀 Starting tunnel for service nginx-deployment.

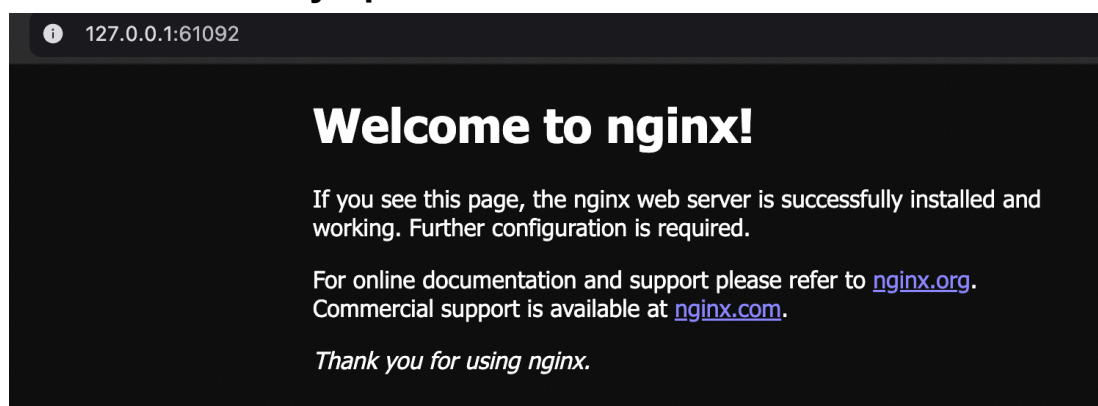
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment		http://127.0.0.1:61071

🎉 Starting tunnel for service nginx-deployment.

🌐 Opening service default/nginx-deployment in default browser...

! Because you are using a Docker driver on darwin, the terminal needs to be open to run it.

This automatically opens the browser.



Step 8 :Clean-up Process

```
kunaltiwari@riot pr8 % kubectl delete pods --all

pod "nginx-deployment-6f9664446b-hlpn2" deleted from default namespace
pod "nginx-deployment-6f9664446b-vdxfw" deleted from default namespace
kunaltiwari@riot pr8 % kubectl delete svc --all

service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
kunaltiwari@riot pr8 % kubectl delete deployments --all

deployment.apps "nginx-deployment" deleted from default namespace
kunaltiwari@riot pr8 %
```